

TESTING AI BASED APPLICATIONS



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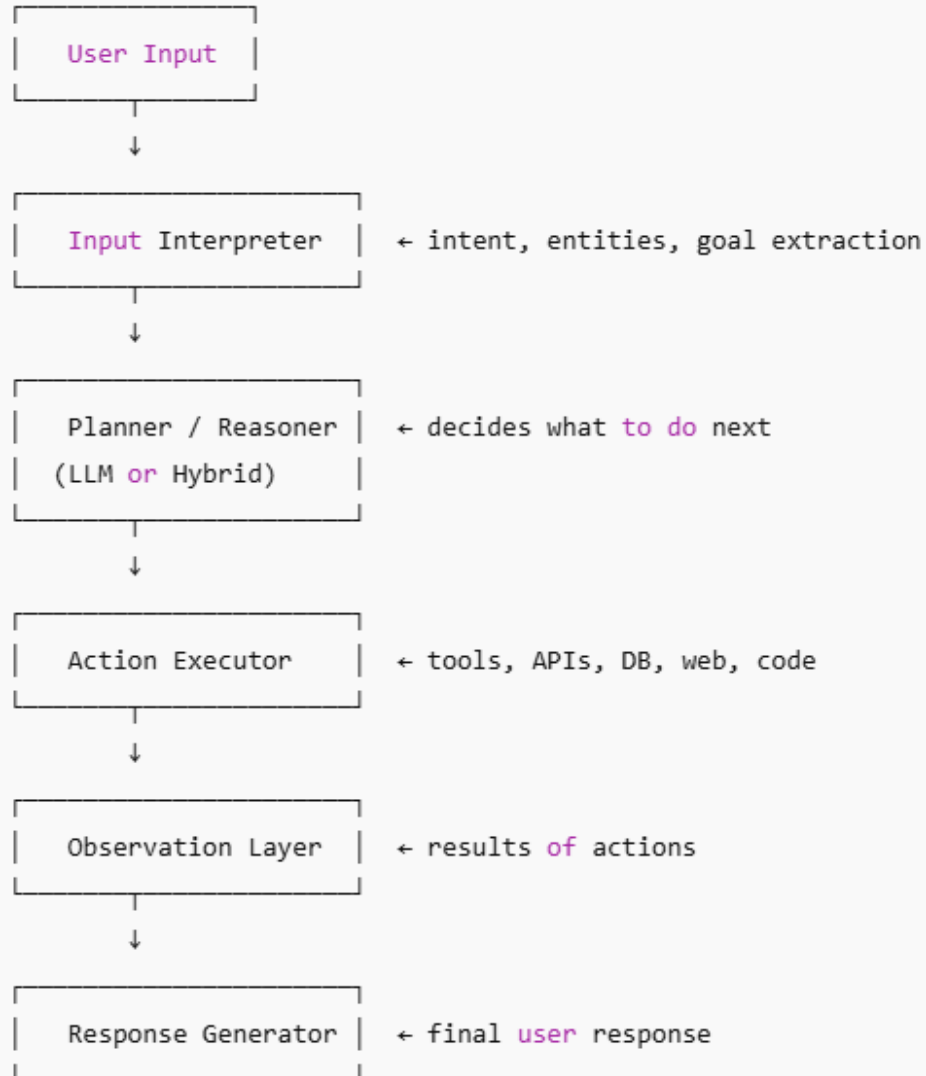


AI AGENT ARCHITECTURE



An LLM plus memory, tools, planning, and feedback loops that can take actions to achieve goals.

MINIMAL AI AGENT ARCHITECTURE



This loop may run
once or many times
until the goal is
achieved.

1. INPUT AND CONTEXT PROCESSOR

"Book a flight to Mumbai tomorrow"

→ intent: book_flight

→ entities: {destination: Mumbai, date: tomorrow}

Purpose: Understand *what the user wants*

2. PLANNER (THE BRAIN)

Example:

1. Search flights
 2. Filter cheapest
 3. Ask user confirmation
 4. Book flight
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2. PLANNER (THE BRAIN)

Decides:

- ☐ What steps are needed
- ☐ Which tools to use
- ☐ When to stop

Types:

- ☐ ReAct (reason + act)
 - ☐ Tree-of-thought
 - ☐ Rule + LLM hybrid
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3. MEMORY

Short-term memory

- Current conversation
- Recent tool outputs

Long-term memory

- User preferences
- Past successful strategies
- Stored embeddings

Example: “**User** prefers morning flights”

4. TOOL / SKILL LAYER

Examples:

- REST APIs
- Databases
- Files Systems
- Internal services

Described to LLM like:

`name`: search_flights

`inputs`: destination, date

`output`: list of flights

This makes the agent **useful**.

5. OBSERVATION & FEEDBACK LOOP

After each action:

- Did it succeed?
- Did it error?
- Do we need to retry?

This is how agents:

- ✓ Self-correct
 - ✓ Handle failures
 - ✓ Adapt dynamically
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6. POLICY & SAFETY GUARD

Runs:

- ☐ Before tool calls
- ☐ Before final response

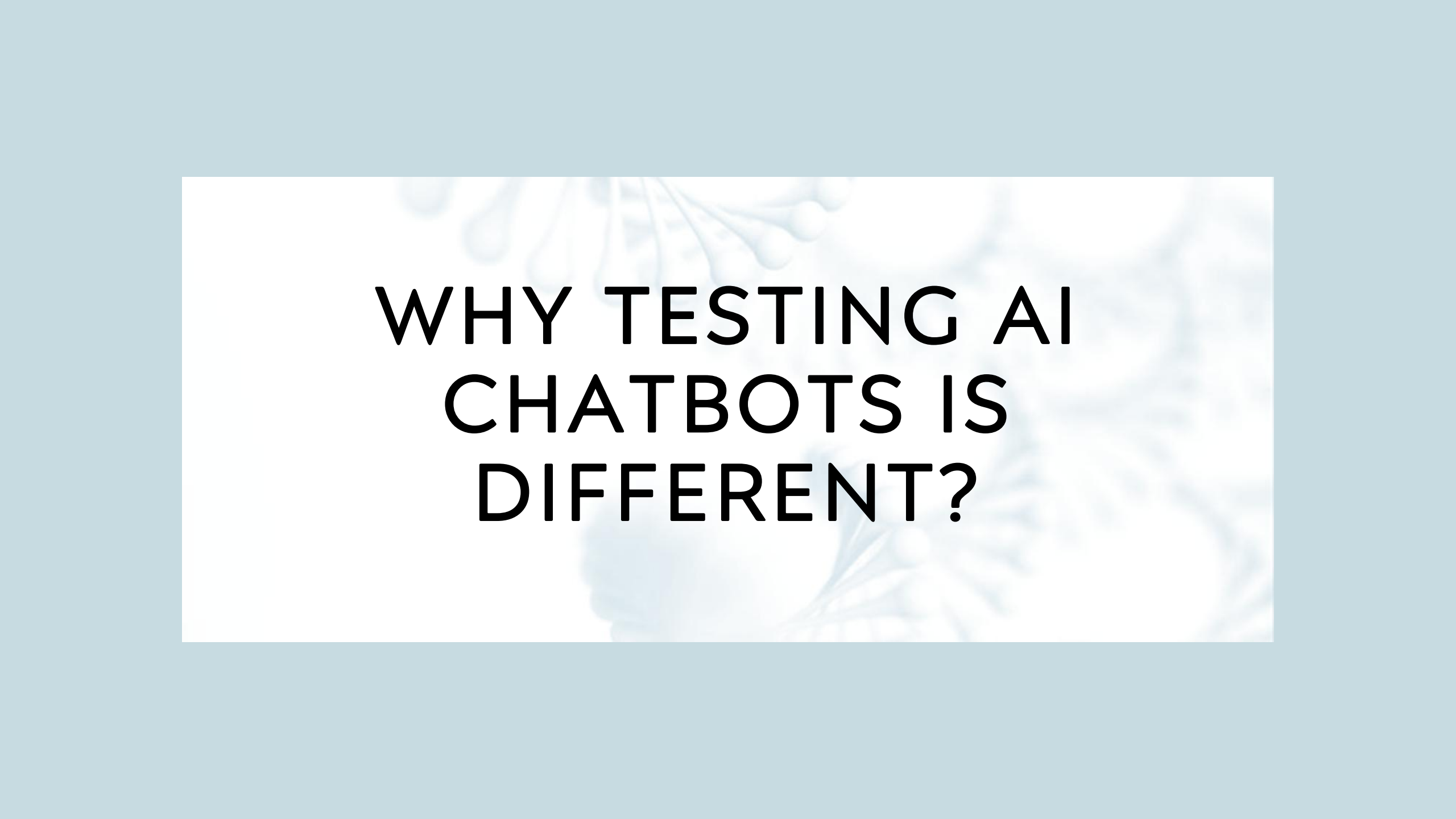
Checks:

- ✓ Allowed actions
 - ✓ Data access
 - ✓ Compliance rules
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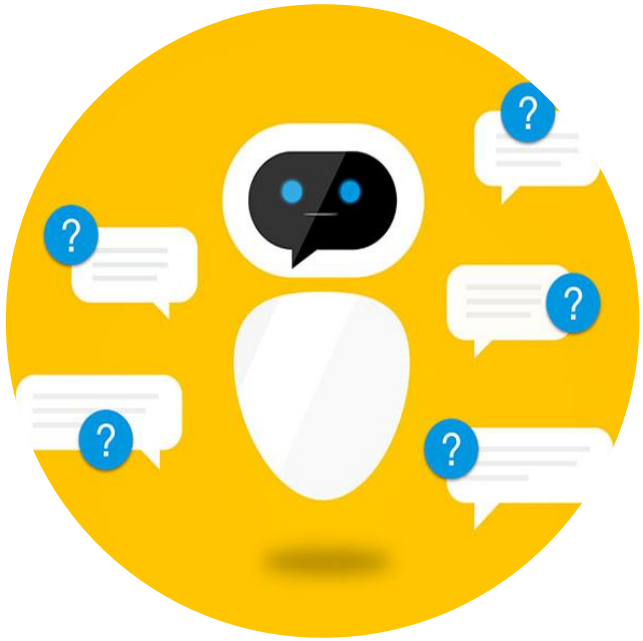
YOU WOULD TEST:

- PLANNER DECISIONS
- TOOL USAGE CORRECTNESS
- MEMORY UPDATES
- SAFETY VIOLATIONS
- END-TO-END GOAL SUCCESS
- BENEFITS



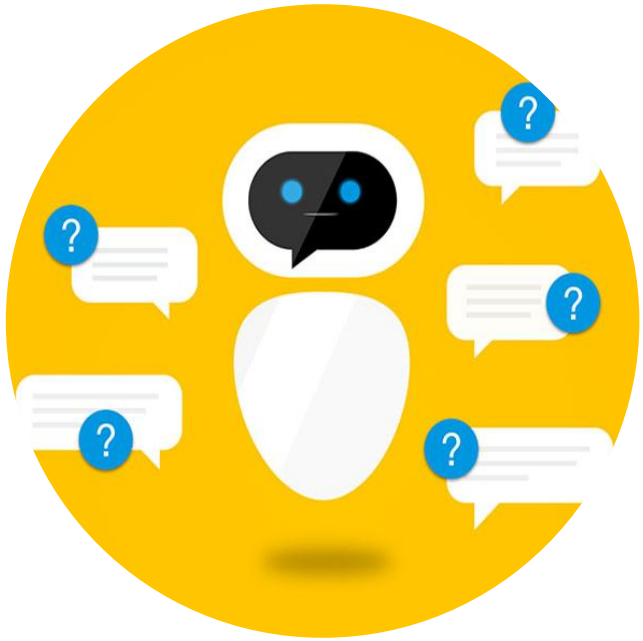


WHY TESTING AI CHATBOTS IS DIFFERENT?



TESTING AI SYSTEMS

- ❑ TRADITIONAL SOFTWARE TESTING VS AI-DRIVEN SYSTEMS
- ❑ DETERMINISTIC VS PROBABILISTIC BEHAVIOUR
- ❑ WHY "EXPECTED OUTPUT" IS NO LONGER A SINGLE VALUE



ROLE OF A TESTER IN AI CHATBOT TEAMS

- ❑ NOT JUST BUG FINDING, BUT RISK IDENTIFICATION
- ❑ PARTNERING WITH DATA SCIENTISTS, ML ENGINEERS, PRODUCT
- ❑ OWNING QUALITY OF CONVERSATIONS, NOT JUST API

TYPICAL AI CHATBOT ARCHITECTURE

Components of a chatbot architecture

- UI (Web / Mobile / Voice)
 - Backend Orchestrator
 - LLM (OpenAI, Azure OpenAI, Claude, etc.)
 - Prompt Layer
 - Tools / Plugins / APIs
 - Vector DB (RAG)
 - Memory / Context Store
 - AI Workers / Agents
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WHERE TESTING FITS

Prompt testing

RAG validation

End-to-end
conversation testing

Model behavior
validation

Tool invocation
testing

TYPES OF TESTING FOR AI CHATBOTS

Only in context with AI, additional to existing testing that we already do as part of traditional testing approach.



FUNCTIONAL TESTING

- INTENT UNDERSTANDING
- ENTITY EXTRACTION
- CORRECT RESPONSE CATEGORY
- TOOL / API SELECTION

EXAMPLE:

USER ASKS: "BOOK ME A FLIGHT TOMORROW"

DID CHATBOT:

- *Understand intent?*
- *Ask missing details?*
- *Call correct service?*

CONVERSATIONAL TESTING

- MULTI-TURN CONVERSATIONS
- CONTEXT CARRYOVER
- FOLLOW-UP QUESTIONS
- TOPIC SWITCHING

EXAMPLE:

- “BOOK A CAB”
- “MAKE IT FOR TOMORROW MORNING”
- “ACTUALLY, CANCEL IT”

NON-FUNCTIONAL TESTING

- PERFORMANCE
(LATENCY OF LLM
RESPONSES)
- SCALABILITY
- RELIABILITY UNDER
LOAD
- COST AWARENESS
(TOKEN USAGE)

AI-SPECIFIC TESTING

- HALLUCINATION
DETECTION
- BIAS & FAIRNESS
CHECKS
- TOXICITY & SAFETY
- PROMPT INJECTION
ATTACKS

TEST DESIGN TECHNIQUES FOR AI CHATBOTS



DESIGNING TEST CASES WITHOUT FIXED EXPECTED RESULTS

- GOLDEN RESPONSE RANGES
- SEMANTIC SIMILARITY INSTEAD OF EXACT MATCH
- INTENT CORRECTNESS OVER WORDING

GOLDEN RESPONSE RANGE

INSTEAD OF ONE SINGLE “PERFECT” ANSWER, TESTERS DEFINE A RANGE OF ACCEPTABLE CORRECT RESPONSES (GOLDEN RESPONSES). IF THE CHATBOT’S REPLY FALLS WITHIN THIS RANGE, IT IS CONSIDERED CORRECT

USER: “WHAT DOCUMENTS ARE NEEDED TO OPEN A BANK ACCOUNT IN INDIA?”

ACCEPTABLE (GOLDEN) RESPONSES MAY INCLUDE:

- AADHAAR AND PAN CARD
- AADHAAR WITH ANY VALID ID PROOF
- PAN CARD, ADDRESS PROOF, AND PHOTOGRAPH

EVEN IF WORDING DIFFERS, ALL ARE VALID AS PER INDIAN BANKING NORMS.

SEMANTIC SIMILARITY INSTEAD OF EXACT MATCH

THE CHATBOT'S
RESPONSE IS
EVALUATED BASED ON
MEANING, NOT EXACT
WORDING.

EXPECTED ANSWER:

“YOU CAN CHECK YOUR AADHAAR
STATUS ON THE UIDAI WEBSITE.”

CHATBOT ANSWER:

“VISIT THE OFFICIAL UIDAI PORTAL
TO TRACK YOUR AADHAAR
APPLICATION.”

*ALTHOUGH WORDS DIFFER, THE
MEANING IS THE SAME, SO THE
RESPONSE IS CORRECT.*

INTENT CORRECTNESS OVER WORDINGS

FOCUS ON WHETHER THE CHATBOT UNDERSTANDS THE USER'S INTENT, EVEN IF THE PHRASING IS INFORMAL, MIXED-LANGUAGE, OR GRAMMATICALLY INCORRECT.

USER:

"I WAS CHARGED TWICE FOR MY UPI PAYMENT."

ACTUAL INTENT:

THE USER WANTS TO **REPORT A FAILED/DUPLICATE TRANSACTION AND SEEK REFUND GUIDANCE**, NOT JUST INFORMATION ABOUT UPI.

CORRECT CHATBOT RESPONSE:

"YOU CAN RAISE A COMPLAINT FOR A DUPLICATE UPI TRANSACTION THROUGH YOUR BANK'S APP OR THE NPCI GRIEVANCE PORTAL. PLEASE KEEP THE TRANSACTION ID AND DATE HANDY."

EVEN THOUGH THE USER DOES NOT EXPLICITLY SAY "COMPLAINT" OR "REFUND," THE CHATBOT CORRECTLY IDENTIFIES THE **INTENT** AND PROVIDES THE RIGHT NEXT STEPS.

TEST DATA STRATEGIES

- ❑ HAPPY PATH CONVERSATIONS
- ❑ EDGE-CASE PROMPTS
- ❑ ADVERSARIAL PROMPTS
- ❑ AMBIGUOUS INPUTS

EVALUATION METRICS

- ❑ INTENT ACCURACY
- ❑ RESPONSE RELEVANCE
- ❑ FACTUAL CORRECTNESS
- ❑ CONVERSATION SUCCESS RATE

Exercise: Explore Ragas <https://docs.ragas.io/en/stable/getstarted/>

RAG – SPECIFIC TEST SCENARIOS

- ANSWER MUST COME FROM PROVIDED DOCUMENTS
- NO HALLUCINATION OUTSIDE KNOWLEDGE BASE
- CORRECT CITATION / REFERENCE



“HOW DO YOU DECIDE IF AN AI CHATBOT
RESPONSE IS CORRECT?”

THANK YOU



"THE ONLY WAY TO DO GREAT WORK IS TO LOVE WHAT YOU DO." — STEVE JOBS 🌟